

Fall 2025 EE582 Homework 02 Report

Nodal-Analysis-based Modeling of Electric Circuits

Aryan Ritwajeet Jha

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Section A: Question 1

One way to analyze the properties of various discretization methods (e.g., in terms of their numerical oscillations) is to solve a lossless first-order system (e.g., inductance or capacitance) when the state variable of the component is suddenly changed. The circuit considers a case where a dc voltage is suddenly applied to a capacitor.

- a) Determine the step-by-step solution for the current in the capacitor using:
 - i) Forward Euler,
 - ii) Backward Euler,
 - iii) Trapezoidal.

Section A: Question 2

Comment on how the simulation time-step size impacts the numerical solution, and which method is preferred in this case.

Section B: Question 1

The parameters of the circuit below are: $R = 10\Omega$, $L = 20mH$, $e(t) = 10Vdc$, and the switch is closed at $t = 0$. The circuit is to be solved from $t = 0$ to $t = 10ms$ using:

- a) PSCAD software,
- b) your own MATLAB code based on the nodal analysis method.